Lab I Guide: Explore the Google Cloud Console and Cloud Shell

Course: Essential Google Cloud Infrastructure: Foundation

Overview

In this lab, you become familiar with the Google Cloud web-based interface. There are two integrated environments: a graphical user interface (GUI) called the Google Cloud console, and a command-line interface (CLI) called Cloud Shell. You use both environments in this lab.
Things to know about the Cloud console:

- The Cloud console is under continuous development, so the layout occasionally changes to accommodate new features, which can mean a slightly different workflow.
- You can perform most common Google Cloud actions in the console, but not all. Very new technologies, or detailed API and command options, may not be available there. In those cases, use the command line or the API.
- The console is very fast for some activities. It can perform multiple actions on your behalf that would otherwise need many CLI commands, and it handles repetitive actions in a few clicks.
- The console reduces errors by offering only valid options in its menus, and it validates configuration behind the scenes before submitting changes. A command line can't do this kind of dynamic validation.

Objectives

- Get access to Google Cloud.
- Use the Cloud console to create a Cloud Storage bucket.
- Use Cloud Shell to create a Cloud Storage bucket.
- Become familiar with Cloud Shell features.

Setup: Sign in to the Google Cloud console

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method.
2. Review the **Lab setup and access** panel on the right. It contains the **Open Google Cloud console** button, the temporary username and password for this lab, and any other information you need. The lab timer near the top of the page shows the remaining time.
3. Click **Open Google Cloud console**. If you are using Chrome, you can right-click and select **Open Link in Incognito Window**. The lab spins up resources and opens another tab with the **Sign in** page.

4. Tip: arrange the lab tab and the console tab in separate windows, side by side.
5. If you see the **Choose an account** dialog, click **Use Another Account**.
6. Copy the **Username** from the Lab setup and access panel, paste it into the Sign in dialog, and click **Next**.
7. Copy the **Password** from the Lab setup and access panel, paste it into the Welcome dialog, and click **Next**.
8. Click through the remaining pages: accept the terms and conditions, do not add recovery options or two-factor authentication (this is a temporary account), and do not sign up for free trials.
9. Wait a few moments for the Google Cloud console to open in this tab.

Important: You must use the credentials the lab provides. Do not use your own Google Cloud account credentials, as using your own account may incur extra charges.
Note: To see a list of Google Cloud products and services, click the **Navigation menu** at the top left, or type the product name in the **Search** field.

Task 1. Create a bucket using the Cloud console

In this task, you create a bucket and become familiar with how actions are presented in the console.

Create the bucket

1. In the Google Cloud console, on the **Navigation menu**, click **Cloud Storage** > **Buckets**.
2. Click **Create**.
3. For **Name**, type a globally unique bucket name. Leave all other values at their defaults.
4. Click **Create**.
5. If prompted that **Public access will be prevented**, click **Confirm**.

Explore console features

The Google Cloud menu contains a **Notifications** icon. Feedback from the underlying commands is sometimes shown there. If you are not sure what is happening, check Notifications for more information and history.

Task 2. Access Cloud Shell

Cloud Shell lets you manage projects and resources from the command line without installing the Cloud SDK or other tools on your computer. Cloud Shell provides:

- A temporary Compute Engine VM
- Command-line access to the instance through a browser
- 5 GB of persistent disk storage (the $HOME directory)
- The Cloud SDK and other tools, pre-installed:
    - **gcloud**: for working with Compute Engine and many Google Cloud services
    - **gcloud storage**: for working with Cloud Storage
    - **kubectl**: for working with Google Kubernetes Engine and Kubernetes
    - **bq**: for working with BigQuery
- Language support for Java, Go, Python, Node.js, PHP and Ruby

- Web preview functionality
- Built-in authorization for access to resources and instances

Note: After 1 hour of inactivity, the Cloud Shell instance is recycled. Only the /home directory persists. Any changes to the system configuration, including environment variables, are lost between sessions.

Open Cloud Shell and explore its features

1. On the Google Cloud console title bar, click **Activate Cloud Shell**. If prompted, click **Continue**. Cloud Shell opens at the bottom of the console window.
2. Find the four buttons on the far right of the Cloud Shell toolbar:
    - **Minimize**: minimizes the window.
    - **Restore/Maximize**: restores the window if minimized, or maximizes it, giving you full access to the console without closing Cloud Shell.
    - **Open in a new window**: pops Cloud Shell out into a full-sized terminal window, useful for editing files or viewing long command output.
    - **Close terminal**: closes Cloud Shell. Every time you close Cloud Shell, the VM is reset and all machine context is lost.
3. Close Cloud Shell.

Check your understanding

Question: Which features and capabilities does Cloud Shell provide? (Select all that apply.)

- A command-line tool that requires you to install Cloud SDK
- Built-in authorization for access to resources and instances
- Command-line access to a free temporary Compute Engine VM
- 5 GB of persistent storage (/home)

Answer: Built-in authorization for access to resources and instances; command-line access to a free temporary Compute Engine VM; and 5 GB of persistent storage (/home). Cloud Shell does not require you to install the Cloud SDK, because it comes pre-installed.

Task 3. Use Cloud Shell to create a Cloud Storage bucket

1. Open Cloud Shell again.
2. Use the gcloud storage command to create another bucket. Replace [BUCKET_NAME] with a globally unique name. You can append a 2 to the bucket name you used in Task 1.

```
gcloud storage buckets create gs://[BUCKET_NAME]
     # use --location=REGION_NAME if needed
```

3. If prompted, click **Authorize**.
4. In the Google Cloud console, on the **Navigation menu**, click **Cloud Storage** > **Buckets**, or click **Refresh** if you are already in the Storage browser.
5. Confirm that the second bucket appears in the Buckets list.

Note: You have performed equivalent actions using the Cloud console and Cloud Shell: one bucket created in each.

Task 4. Explore more Cloud Shell features

Upload a file

1. Open Cloud Shell.
2. Click the **More** button in the Cloud Shell toolbar to display further options.
3. Click **Upload**, and select a file from your local machine to upload to the Cloud Shell VM. This guide calls it [MY_FILE].
4. In Cloud Shell, run the following command to confirm that the file was uploaded:

```
ls
```

5. Copy the file into one of the buckets you created. Replace [MY_FILE] with your file name and [BUCKET_NAME] with one of your bucket names:

```
gcloud storage cp [MY_FILE] gs://[BUCKET_NAME]
```

If your file name contains spaces, put single quotes around it, for example:

```
gcloud storage cp 'my file.txt' gs://[BUCKET_NAME]
```

6. Explore the options available in Cloud Shell by clicking the different icons in the toolbar.
7. Close all Cloud Shell sessions.

Note: You have uploaded a file to the Cloud Shell VM and copied it to a bucket.